Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

A printed claim that the scorecaster breaks online conformal coverage meets its refutation by overlapping authors; this paper joins them and locates the edge of the set that refutation relies on. Conformal PID states long-run coverage for any bounded scorecaster; a corollary keeps it for predictable perturbations of the deployed threshold inside an admissible radius. The claim was acted on; the readout's placement and the corollary's derivation are conceded by name. At the null scorecaster that condition is *tight*. A legal dead band on the *completed* threshold leaves that set at its radius, and past it miscoverage is 1.000000 against $\alpha = 0.1$: coverage fails outright, not merely its rate. The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere the condition is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

1 Introduction

A refereed paper states that a scorecaster breaks Conformal PID's coverage guarantee. A later one, by the same first and last authors, proves in the same notation that a bounded perturbation of the deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the joining is what this paper adds.

An online conformal method [Angelopoulos and Bates, 2023] moves a threshold each round, and that movement is not an unnamed quantity. Forecast stability names, scores and prices it [Tunc et al., 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Holding the level fixed and comparing the temporal structure came earlier still, in forecast verification, which matches predictive marginals on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. A third strand reports that the two numbers usually printed of a deployed interval, coverage and mean length, under-describe it [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

The record carries a claim and its refutation; this paper joins them. Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be predictable and lie in $[-b/2, b/2]$: "any function of the past" (Theorem 1). Such a readout is already quantified over. The claim reads "[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee" [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, verbatim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2] add any predictable d_t, $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ with $\mu_t \in [0, 1]$ predictable, to the deployed threshold; since $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result "has exactly the CPID error-integration form" and Proposition 2's own finite-horizon bound on $|E_T|$ is retained, whence $|E_T| = o(T)$. The same generic-slot argument carries AcMCP's long-run coverage [Wang and Hyndman, 2024]. Four surfaces were searched for both identifiers and "scorecaster". They are arXiv full text, arXiv abstracts, OpenReview, and the six works Semantic Scholar records citing

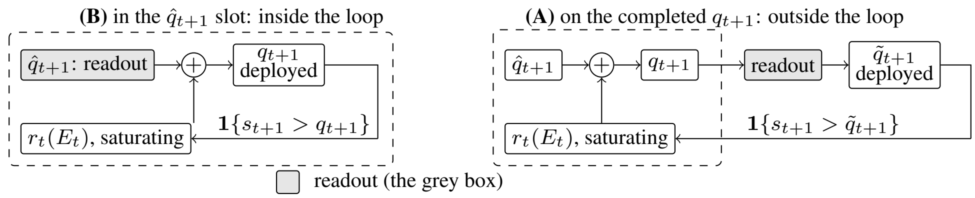

Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In **(B)** the loop closes on the integrator's own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range, inside Corollary 2's admissible set. In **(A)** it closes on $\tilde{q}_{t+1}$, which it never computed, and nothing keeps the deployed sequence inside that set.

Li and Rodríguez [2025]. None returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice "arbitrary" and lacking "principled guidance" while asserting valid coverage: model selection, not validity. **We claim neither the placement nor the derivation.**

Where Corollary 2 stops, its radius condition is tight. It is a *retention* result, silent outside that radius: perturbations inside the radius keep the guarantee, and perturbations outside it are not addressed either way. A sufficient condition can be conservative, so the question the corollary leaves open is whether its radius is sharp, that is, whether something legal just outside it actually fails. Section 3 exhibits a legal perturbation leaving it: the L_1 dead-band family characterised here on the *completed* threshold, whose edge coincides with the corollary's radius at the null scorecaster. Past it coverage fails outright, not merely its rate: the published sufficient condition is necessary there. Section 4 sets both claims beside four vocabularies for the same movement.

2 Setup

The producer. At time t a forecaster fit on data before t induces a score s_t, and a threshold q_t is emitted before y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \le q_t\}$ and $\mathrm{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The target is long-run coverage, $T^{-1}\sum_{t\le T}\mathrm{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score scale, generalizing the running-error feedback that adaptive conformal inference [Gibbs and Candès, 2021] and its distribution-shift variant DtACI [Gibbs and Candès, 2024] already apply directly to the threshold:

$$q_{t+1} \;=\; \hat{q}_{t+1} \;+\; r_t\Big(\textstyle\sum_{i\le t}(\mathrm{err}_i - \alpha)\Big), \tag{1}$$

Angelopoulos et al.'s iteration (5). Its r_t saturates: condition (4) requires $r_t(x) \ge b$ for $x \ge c\,h(t)$ and $r_t(x) \le -b$ for $x \le -c\,h(t)$, with h nonnegative, nondecreasing and sublinear. In words: once the accumulated error clears $c\,h(t)$ in either direction, the correction is at least a full b in the direction that reduces it. With $E_t = \sum_{i\le t}(\mathrm{err}_i - \alpha)$, Proposition 2 gives $|E_T| \le c\,h(T) + 1$, read on realised scores; that is a coverage gap of at most $(c\,h(T)+1)/T$, which vanishes because h is sublinear.

One scalar, two readouts, two places. Damping a sequentially updated scalar has two standard readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1-w)\hat{q}^{\mathrm{raw}}_{t+1} + w\tilde{q}_t$ [Gârleanu and Pedersen, 2013], which Godahewa et al. [2025] publish as post-processing and Binny and Dixit [2025, Eq. 13] apply to a deployed conformal threshold. A proportional cost gives a no-trade band, moving only when the target is more than τ away, then pulling back exactly τ [Constantinides, 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1. **(A)**, Placement A, is the one measured here, and it is the one a designer reaches for, because the deployed number is what has to sit still. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form*; it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits Proposition 2's rate while covering, L_1 past the radius loses coverage, and at the legal $\hat{q} \equiv -b/2$ both lose it. So the tight quantity is the radius, not either form.

The quantity, and the names it already has. We report the deployed threshold's *travel*, $\mathcal{T}_H = \sum_{t=2}^{H}|q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new, and the name is a convenience rather than a contribution. Over a predictive distribution $\sum|\Delta q|$ is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls the same movement *jumpiness* [Zsótér et al., 2009], a *convergence index* [Ehret, 2010] and forecast *(in)consistency* [Pappenberger et al., 2011], all three imported by Van Belle et al. [2026]; applied control calls it IACER [Sarhadi, 2025]; and dynamic-regret online learning calls it a path length. Section 4 names ten across four vocabularies, so a fresh one is used here, for one reading only.

Harness. The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b\,\mathrm{clip}(x/(c\,h(t)), -1, 1)$, with $c = 1$ and $h(t) = \log(t+2)$, meets (4) with equality. The scorecaster is null and legal, reducing (1) to the pure error integration Proposition 2 addresses, so the readout is the only manipulated variable. Neither choice is neutral, and each sits at an edge of what condition (4) and Theorem 1 allow rather than in the middle of it: (4) is a lower bound, and this saturator meets it exactly rather than exceeding it, while Theorem 1 permits scorecasters anywhere within $b/2$ either side of null, predictable but not required to be constant in t. The adversary is specified, not tie-broken. It plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the $\varepsilon \downarrow 0$ reading of "sets the score at the threshold", firing the strict indicator. Legality is Theorem 1's $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight. *That specification names one member of a class*: any adapted $s_t \in [-b/2, b/2]$ is legal, the constant $s_t \equiv -b/2$ included. Every number in Table 1 is measured against the one adversary above; Section 3's two boundaries are quantified over the class, and they differ.

3 The edge of the admissible set, located and proved

The retention result is published, and tested from outside. Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \le \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration form and *retains* $|E_T| \le c\,h(T) + 1$: the radius is theirs. A downstream partial adjustment of weight w leaves it; Table 1 prices that departure, and five of its eleven arms exceed the bound at $T = 10^6$, and for $w = 0.99$ and 0.999 the excursion is *flat in T* while the bound grows like $\log T$. The forfeit is a property of the smoothing *strength*, not Placement A, and it buys something: travel falls from 28,311 to 202. The forfeit's sign is counter-intuitive, so it is worth saying what does and does not break. A readout on the completed q_{t+1} touches neither r_t nor the integrator's argument, so it cannot put (4) at risk. What it does instead is delay a correction that has already been computed, so the accumulator excurses *further* rather than less: smoothing the output does not damp E_t but feeds it.

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2's bound is 13.2061 at $T = 2\times10^5$ and 14.8155 at $T = 10^6$; "ratio" is $\max_t |E_t|$ over it, and "Sat." the fraction of rounds on which (4) delivers its full $\pm b$. Every number is measured against the adversary Section 2 specifies.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t \lvert E_t\rvert$ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	**0.100004**	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	**1.000000**	**1.000000**	900,000	60,747	1.0000	0.5

The L_1 dead band leaves the radius, and the edge is a theorem. A dead band pulls the deployed threshold back by exactly τ when it releases and holds it otherwise, so $|\tilde{q}_{t+1} - q_{t+1}| \le \tau$ at every round and "inside Corollary 2" reads $\tau \le \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator's ceiling, and the deployed value's realised supremum is $b - \tau$ to the printed digit at every $\tau \ge 0.9$. Outside the radius the answer turns on two different levels of the integrator, and condition (4) names only one of them. The two are the largest correction r_t ever issues and the correction it is guaranteed to issue once the accumulator has cleared the radius; where they differ, a boundary written with the first of them over-certifies. Write $A_t^+ = \sup_x r_t(x)$ and $A_t^- = \inf_x r_t(x)$ for the integrator's extremes, and $\Lambda_t^+ = \inf_{x \ge c\,h(t)} r_t(x)$, $\Lambda_t^- = \sup_{x \le -c\,h(t)} r_t(x)$ for the levels it is *guaranteed* to reach once the accumulator clears the radius. Condition (4) says exactly $\Lambda_t^+ \ge b$ and $\Lambda_t^- \le -b$; that $\Lambda_t^\pm = A_t^\pm$, that the integrator reaches its extremes where (4) puts them, is a further hypothesis and not one (4) supplies. Set

$$\tau^{\star\pm} \;=\; \sup_t \big(|A_t^{\pm}| \pm \hat q_{t+1}\big) - b/2, \qquad \sigma^{\pm} \;=\; \inf_t \big(|\Lambda_t^{\pm}| \pm \hat q_{t+1}\big) - b/2. \tag{2}$$

Failure, over the whole admissible class. If $\tau > \tau^{\star+}$ then $\hat q_{t+1} + A_t^{+} - \tau < b/2$ at every round, so $\tilde q_{t+1} \le \max(\tilde q_t, q_{t+1} - \tau)$ makes $\{\tilde q < b/2\}$ absorbing: a release upward lands below $b/2$, and a hold or a release downward cannot lift the deployed value at all. Against $s_t = \mathrm{clip}(\tilde q_t + \varepsilon, \pm b/2)$ the indicator is then $\mathbf{1}\{\tilde q_t < b/2\}$ and every round misses. If $\tau \ge \tau^{\star-}$ the constant legal play $s_t \equiv -b/2$ makes $\{\tilde q \ge -b/2\}$ absorbing by the mirror inequality, so no round can miss and the realised rate is 0 rather than α. Either way long-run coverage is lost, with $|E_T|$ linear in T. Nothing here asks $\hat q$ to be constant, asks the extremes to be attained anywhere, or asks $A_t^{\pm}$ to be finite beyond what the hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If $\tau \le \sigma^{+}$ then $E_{t-1} \ge c\,h(t-1)$ forces $q_t \ge b/2 + \tau$ and so $\tilde q_t \ge b/2$, whence $\mathrm{err}_t = 0$ against *every* legal adversary; if $\tau < \sigma^{-}$ the mirror forces $\mathrm{err}_t = 1$ below $-c\,h(t-1)$. The accumulator cannot then leave $[-c\,h(t), c\,h(t)]$ by more than one round, and $|E_T| \le c\,h(T) + 1$: Proposition 2's own bound, retained verbatim rather than weakened to $o(T)$. Using condition (4) alone this already certifies $\tau \le b/2 + \inf_t \hat q_t$ and $\tau < b/2 - \sup_t \hat q_t$, which for a constant scorecaster is Corollary 2's radius at $\mu_t = 1$.

The two directions meet, with no gap, on the sub-class the five settings inhabit. If $\hat q_t \equiv \hat q$, if $\Lambda_t^{\pm} = A_t^{\pm}$, and if neither depends on t, then $\sigma^{\pm} = \tau^{\star\pm}$ and **coverage is retained against every legal adapted adversary if and only if $\tau \le \tau^{\star+}$ and $\tau < \tau^{\star-}$.** For a symmetric saturator that is one line: coverage if and only if $\tau \le \sup_x r_t(x) - b/2 - |\hat q|$, with the endpoint included only when $\hat q < 0$. Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed before is the failure half.** $\tau^{\star+}$ is the rule Figures 2 and 3 draw, and all 19 dead-band runs agree with it read against the adversary the harness plays. It is not a retention condition, and Table 2 (App. A) separates the two with legal objects only. The supremum can sit where the dynamics never go: $|E_t| \le (1-\alpha)t$, so a saturator equal to the harness's inside $|x| \le t^2$ and to $\pm 10b$ outside it satisfies (4), reads $\tau^{\star+} = 19$, and fails at $\tau = 1.001$. A legal time-varying $\hat q$ splits $\sup_t$ from $\inf_t$: $\hat q_t = b/2$ on t a power of two and 0 elsewhere reads $\tau^{\star+} = 2$ against $\sigma^{+} = 1$. And that printed rule omits $\tau^{\star-}$ entirely.

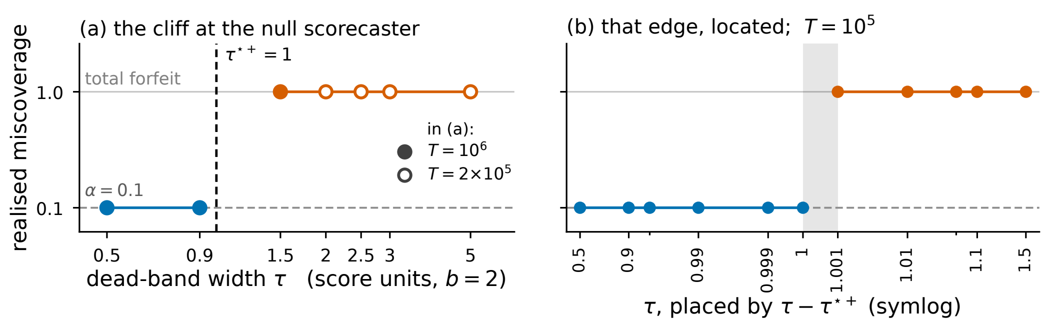

Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat q \equiv 0$ with a saturator of level b, under the *specified* adversary, where the upward boundary $\tau^{\star+}$ of (2) is 1. **(a)** The full sweep: filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2{\times}10^5$; blue covers, orange forfeits, and the dashed rule is $\tau^{\star+}$. **(b)** The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^{\star+}$ on a symlog axis whose linear zone is $|\tau - \tau^{\star+}| \le 10^{-3}$, which magnifies the neighbourhood of $\tau^{\star+}$: visual spacing here is not proportional to true distance in τ. Unlabelled minor ticks mark $\tau = 0.95$ and 1.05. The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four $(r_t, \hat q)$ settings. The same eleven widths under the *mirror* adversary, which locates $\tau^{\star-}$, are this run's exact-threshold column read under the strict indicator.

The sharp form is one number, and its hypotheses are what buy it. Specialise to a symmetric saturator that meets (4) with equality and attains its extremes there ($\Lambda_t^{\pm} = A_t^{\pm} = \pm b$; the harness's $r_t = b\,\mathrm{clip}(x/(c\,h(t)), \pm 1)$ is one, Section 2), and to a constant $\hat q$. Then

$$\min(\tau^{\star+}, \tau^{\star-}) \;=\; b/2 - |\hat q|, \tag{3}$$

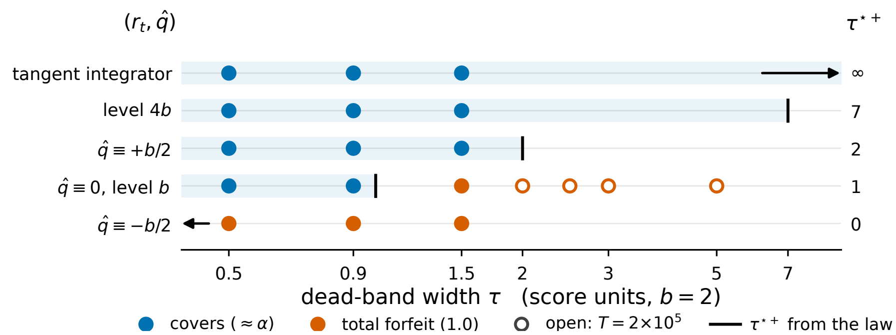

Figure 3: All 19 dead-band runs, at the five admissible $(r_t, \hat{q})$ settings: rows are settings ordered by $\tau^\star$, and the shared horizontal axis is τ. Blue markers cover at $\approx \alpha$ and orange markers return 1.000000; filled markers run to $T = 10^6$ and open markers to $T = 2{\times}10^5$, all under the *specified* adversary. The rule on each row is that setting's upward boundary $\tau^{\star+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, and the tinted band is $\tau \le \tau^{\star+}$. Two rules fall off the axis and are drawn as arrows: $\tau^{\star+} = 0$ at $\hat{q} \equiv -b/2$, and $\tau^{\star+}$ unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary $\tau^{\star-}$ of (2) is not drawn here and differs from $\tau^{\star+}$ at the two non-null scorecasters.

with the endpoint admissible only when $\hat{q} < 0$: exactly Corollary 2's radius at $\mu_t = 1$, now as an if-and-only-if. **The necessity direction above assumes none of it**: $\tau > \tau^{\star+}$, or $\tau \ge \tau^{\star-}$, forfeits coverage against every admissible r_t and every legal predictable $\{\hat{q}_t\}$, extremes attained or not, and that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather than decorative: the unattained supremum and the varying-$\hat{q}$ split two paragraphs above are exactly what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they always do.

The mirror boundary is already measured, on this paper's own grid. The τ-cliff run's exact-threshold column under the strict indicator is the mirror adversary $s_t \equiv -b/2$ (one error process, $\mathrm{err}_t = \mathbf{1}\{\tilde{q}_t < -b/2\}$), and on the same eleven widths at the null scorecaster it locates $\tau^{\star-} = 1$, closed on the failing side, exactly where (2) puts it. Both 1.000000 and 0.000000 are failures of the same order, $|E_T| = (1-\alpha)T$ and αT, **so the failure criterion is $|E_T|$, not the miscoverage rate**; and once $\hat{q}$ varies the rate stops being a tell at all.

Both degenerate edges, and what they need. An unbounded supremum is not on its own enough for $\tau^{\star+} = \infty$ to mean what it looks like: the same unreachable-set construction with $r_t(x) = x$ past $|x| > t^2$ has no finite supremum and fails at $\tau = 1.5$. What the tangent family has, and that construction does not, is a genuine reach: $\Lambda_t(X) = \tan(\arctan(b)X/(c\,h(t)))$, so the accumulator level at which the deployed threshold is forced to $b/2$ is $\kappa(\tau)\,c\,h(t)$ with $\kappa(\tau) = \arctan(b/2 + \tau - \hat{q})/\arctan(b)$, finite at every finite τ. Coverage is retained at every width, as measured; but $\kappa(\tau) > 1$ once $\tau > b/2 + \hat{q}$, so Proposition 2's own *constant* is forfeited past $\tau = b/2$ while coverage is not. At $\tau = 1.5$, $T = 10^6$ the excursion is 15.10 against Proposition 2's 14.8155 and against $\kappa(1.5)\,c\,h(T) + 1 = 15.8530$. At the other edge, $\sup_t \hat{q}_t = -b/2$ forces $\hat{q} \equiv -b/2$, since every $\hat{q}_t$ is at least $-b/2$ already. There $q_{t+1} - \tau \le -b/2 + b - \tau < b/2$ for every $\tau > 0$ and every t, and $\tilde{q}_1 = 0 < b/2$, so the failing set is absorbing from round 1 with no transient: the admissible window is the single point $\tau = 0$, and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$.

What that establishes is that the published condition is tight. At $\mu_t = 1$, $\hat{q} \equiv 0$ the boundary is exactly Corollary 2's radius, crossing it costs coverage rather than the rate ($\tau = 0.9$ covers at 0.100012, $\tau = 1.5$ returns 1.000000 with $\max_t |E_t| = 900{,}000$, $60{,}747$ times Proposition 2's bound at $T = 10^6$; Table 1), and (2) makes that an if-and-only-if against the whole legal adversary class rather than a pattern across five settings. **The claim is that necessity: a tightness result about someone else's sufficient condition**, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere it stays sufficient only, and now provably so: under the tangent integrator Corollary 2 permits $\tau \le b/2$ while no finite width forfeits coverage against any legal adversary, because the reach is genuine at every level. At $\hat{q} \equiv +b/2$ the looseness goes the other way and disappears: Corollary 2 permits radius 0, and (2) permits no width at all against the mirror adversary, the unsmoothed control included, though $\tau = 1.5$ covers there against the specified one. The condition therefore binds at some legal settings and is slack at others, and that asymmetry is what makes "tight" a located claim rather than a global one: what is stated here is where the radius is sharp, and equally where it is not.

The constructive reading, and it is the corollary's rather than ours. Placement B keeps $\{\hat{q}_t\}$ inside the admissible radius, so Theorem 1 carries it already. What Section 3 adds is that the band has two edges, both computable before deployment from the integrator's guaranteed reach and the scorecaster's range, that crossing either is not a graceful degradation, and that between $\sigma^{\pm}$ and $\tau^{\star\pm}$, which separate only when $\hat{q}$ varies in t, neither direction is settled here.

The boundary read as a bet. The boundary is stated in $|E_T|$, which is the net gain of a Skeptic who pays α each round for err_t, so it has a betting reading. Impose, for that reading alone and on nothing else here, the null that $\mathrm{err}_t \mid \mathcal{F}_{t-1}$ is Bernoulli(α), with $\mathcal{F}_{t-1}$ the history through round $t-1$: past the edge an anytime-valid test of the deployed intervals' calibration convicts the $\tau = 1.5$ arm on its fifth interval at the 5% level, and inside the edge the same test cannot reject at that level at any horizon. It adds a diagnostic and a reading, not a result: the test is Ville's inequality applied to the standard martingale $M_t = \prod_{i \le t} \big(1 + \lambda(\mathrm{err}_i - \alpha)\big)$, whose logarithm depends on the path only through t and E_t [Shafer and Vovk, 2019, Ramdas et al., 2023, Vovk and Wang, 2021]; λ is fixed in $[-1/(1-\alpha),\, 1/\alpha]$, and its sign above is chosen after the fact, not pre-registered; M_t is an e-value for that imposed null alone, the accumulator itself is signed and is neither an e-value nor an e-process, and no result above assumes that null or is certified by that test.

4 Where this sits

One quantity, four names. Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. Online conformal prediction gives distribution-free long-run coverage for any bounded scorecaster [Angelopoulos et al., 2023, Thm 1, Prop. 2]. Forecast stability treats the movement as a functional, read out as post-processing [Godahewa et al., 2025, Eq. 3], against a priced decision to switch [Genov et al., 2026, §4.2], [Van Belle et al., 2026]. Verification work uses a matched-level design against real producers from 2012 [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning gives worst-case theory for the penalised problem [Andrew et al., 2013, Thm 2].

The object that settles a deployed interval's validity (the admissible set around the deployed threshold) sits in the first of them. Those four vocabularies are where this paper's claim and its concessions have to be placed. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves; this paper proves that admissible set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID's own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t \le T} q_t \le O(\alpha\, h(T))$ for a sublinear h, supported with a plot rather than derived. Verification's matched-level design tests real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a Thm. 7 trade-off in which sublinear regret is bought by growing θ with T, which grows the ratio [Andrew et al., 2013, Thm. 7].

5 Limitations

The result above is narrow in several specific ways. They are listed here rather than left to be found, and the run-in bold marks where one concession ends and the next begins.

The retention direction is proved with no gap only where the two boundaries meet: a constant scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run here; a genuinely time-varying $\hat{q}$ leaves the band $\sigma^+ < \tau \leq \tau^{\star +}$ open in neither direction (Section 3). And the excursion is specific to the saturator's *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, the certified gap $(c\,h(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75\,b$. Under i.i.d. scores it returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly.

What follows concerns the evidence rather than the statement. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested**: the argument that it stays inside the admissible radius is Theorem 1's, and no arm run here exercises it. **The i.i.d. check above used one seed** (20260820, $T = 10^6$); **the primary, adversarial harness uses none**, and **only its null-scorecaster edge is bracketed**; the others are one-sided.

What remains open is the band between the two boundaries. For a genuinely time-varying $\hat{q}$, coverage in $\sigma^+ < \tau \leq \tau^{\star +}$ is neither proved nor falsified; Table 2's power-of-two rows sit there, and closing that band is future work, not a sweep this paper has already run.

References

Lachlan L. H. Andrew, Siddharth Barman, Katrina Ligett, Minghong Lin, Adam Meyerson, Alan Roytman, and Adam Wierman. A tale of two metrics: Simultaneous bounds on competitiveness and regret. In *Proceedings of the 26th Conference on Learning Theory (COLT 2013)*, pages 741–763, 2013.

Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023. doi: 10.1561/2200000101.

Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. doi: 10.52202/075280-1000.

Allen Emmanuel Binny and Anushri Dixit. Who moved my distribution? Conformal prediction for interactive multi-agent systems. arXiv:2511.11567v1, 2025.

Jiadong Chen, Yang Luo, Xiuqi Huang, Fuxin Jiang, Yangguang Shi, Tieying Zhang, and Xiaofeng Gao. IPOC: An adaptive interval prediction model based on online chasing and conformal inference for large-scale systems. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 202–212, 2023. doi: 10.1145/3580305.3599396.

George M. Constantinides. Capital market equilibrium with transaction costs. *Journal of Political Economy*, 94(4):842–862, 1986. doi: 10.1086/261410.

M. H. A. Davis and A. R. Norman. Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4):676–713, 1990. doi: 10.1287/moor.15.4.676.

Ricarda Duerst, Jonas Schöley, Julia Hellstrand, and Mikko Myrskylä. Calibrating probabilistic forecast paths on past forecast errors: an application to the Finnish total fertility rate. Technical Report WP-2024-016, Max Planck Institute for Demographic Research, 7 2024.

Théo Dupuy, Binbin Xu, Stéphane Perrey, Jacky Montmain, and Abdelhak Imoussaten. Relevance-aware thresholding in online conformal prediction for time series. In *Artificial Intelligence for Healthcare, and Hybrid Models for Coupling Deductive and Inductive Reasoning: First International Joint Conference, HC@AIxIA+HYDRA 2025, Bologna, Italy, October 25–26, 2025, Proceedings*, volume 2830 of *Communications in Computer and Information Science*, pages 196–217, 2026. doi: 10.1007/978-3-032-16708-8_17.

Uwe Ehret. Convergence index: a new performance measure for the temporal stability of operational rainfall forecasts. *Meteorologische Zeitschrift*, 19(5):441–451, 2010. doi: 10.1127/0941-2948/2010/0480.

Sergio Galeani, Sophie Tarbouriech, Matthew Turner, and Luca Zaccarian. A tutorial on modern anti-windup design. *European Journal of Control*, 15(3-4):418–440, 2009. doi: 10.3166/ejc.15.418-440.

Ben Gao, Jordan Patracone, Stéphane Chrétien, and Olivier Alata. Conformal online learning of deep Koopman linear embeddings. arXiv:2511.12760v2, 2025.

Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340, 2013. doi: 10.1111/jofi.12080.

Evgenii Genov, Julian Ruddick, Christoph Bergmeir, Majid Vafaeipour, Thierry Coosemans, Salvador García, and Maarten Messagie. Balancing forecast accuracy and switching costs in online optimization of energy management systems. *Expert Systems with Applications*, 298:129305, 2026. doi: 10.1016/j.eswa.2025.129305.

Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021.

Isaac Gibbs and Emmanuel J. Candès. Conformal inference for online prediction with arbitrary distribution shifts. *Journal of Machine Learning Research*, 25(162):1–36, 2024.

Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(2): 243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

Rakshitha Godahewa, Christoph Bergmeir, Zeynep Erkin Baz, Chengjun Zhu, Zhangdi Song, Salvador García, and Dario Benavides. On forecast stability. *International Journal of Forecasting*, 41(4): 1539–1558, 2025. doi: 10.1016/j.ijforecast.2025.01.006.

Dongjian Hu, Junxi Wu, Shu-Tao Xia, and Changliang Zou. Distribution-informed online conformal prediction. arXiv:2512.07770v2, 2026.

Ankit Lade, Sai Krishna J., and Indar Kumar. Bias-corrected adaptive conformal inference for multi-horizon time series forecasting. arXiv:2604.13253v1, 2026.

Ruipu Li and Alexander Rodríguez. Neural conformal control for time series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 18439–18447, 2025. doi: 10.1609/aaai.v39i17.34029.

Ruipu Li, Daniel Menacho, and Alexander Rodríguez. Optimization-based online conformal prediction for multi-step forecasting. arXiv:2508.13362v3, 2025.

Yizhou Min, Yizhou Lu, Lanqi Li, Zhen Zhang, and Jiaye Teng. Questioning the coverage-length metric in conformal prediction: When shorter intervals are not better. arXiv:2601.21455v2, 2026.

F. Pappenberger, H. L. Cloke, A. Persson, and D. Demeritt. HESS Opinions "on forecast (in)consistency in a hydro-meteorological chain: curse or blessing?". *Hydrology and Earth System Sciences*, 15(7):2391–2400, 2011. doi: 10.5194/hess-15-2391-2011.

P. Pinson and R. Girard. Evaluating the quality of scenarios of short-term wind power generation. *Applied Energy*, 96:12–20, 2012. doi: 10.1016/j.apenergy.2011.11.004.

Kandrika Pritularga and Nikolaos Kourentzes. Forecast congruence: A quantity to align forecasts and inventory decisions. SSRN working paper 4711817, 2024.

Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023. doi: 10.1214/23-STS894.

Pouria Sarhadi. On the standard performance criteria for applied control design: PID, MPC or machine learning controller? arXiv:2503.14379v3, 2025.

Glenn Shafer and Vladimir Vovk. *Game-Theoretic Foundations for Probability and Finance*. Wiley Series in Probability and Statistics. Wiley, 2019. ISBN 9780470903056. doi: 10.1002/9781118548035.

Huseyin Tunc, Onur A. Kilic, S. Armagan Tarim, and Burak Eksioglu. A simple approach for assessing the cost of system nervousness. *International Journal of Production Economics*, 141(2): 619–625, 2013. doi: 10.1016/j.ijpe.2012.09.022.

Jente Van Belle, Ruben Crevits, and Wouter Verbeke. Improving forecast stability using deep learning. *International Journal of Forecasting*, 39(3):1333–1350, 2023. doi: 10.1016/j.ijforecast.2022.06.007.

Jente Van Belle, Honglin Wen, Wouter Verbeke, and Pierre Pinson. Stabilizing distribution-free probabilistic forecasts. arXiv:2605.28531v1, 2026.

Rahul Vaze. Simultaneous coverage and efficiency guarantee in online conformal prediction. arXiv:2607.26577v1, 2026.

Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021. doi: 10.1214/20-AOS2020.

Xiaoqian Wang and Rob J. Hyndman. Online conformal inference for multi-step time series forecasting. arXiv:2410.13115v2, 2024.

Rochelle P. Worsnop, Michael Scheuerer, Thomas M. Hamill, and Julie K. Lundquist. Generating wind power scenarios for probabilistic ramp event prediction using multivariate statistical post-processing. *Wind Energy Science*, 3(1):371–393, 2018. doi: 10.5194/wes-3-371-2018.

Junxi Wu, Dongjian Hu, Yajie Bao, Shu-Tao Xia, and Changliang Zou. Error-quantified conformal inference for time series. arXiv:2502.00818v2, 2025.

Marco Zanotti. Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems. arXiv:2506.05776v3, 2026.

Mingyi Zheng, Hongyu Jiang, Yizhou Lu, and Jiaye Teng. Smoothing-based conformal prediction for balancing efficiency and interpretability. arXiv:2509.22529v1, 2025.

Ervin Zsótér, Roberto Buizza, and David Richardson. "Jumpiness" of the ECMWF and Met Office EPS control and ensemble-mean forecasts. *Monthly Weather Review*, 137(11):3823–3836, 2009. doi: 10.1175/2009MWR2960.1.

A Supporting measurements

Table 2: Legal $(r_t, \hat{q})$ configurations on which the printed $\tau^\star$ over-certifies, each under two legal adversaries. "Proved window" is where (2) certifies retention against every legal adversary; between it and the failure boundaries $\tau^{\star\pm}$ neither direction is claimed, and the rows built on a supremum the dynamics cannot reach or never attain sit there. Here $\emptyset$ marks a configuration for which no width is certified, $\tau = 0$ included. $T = 10^5$ except the two power-of-two rows, at $T = 10^6$. The last row carries no dead band at all.

				$s_t = \tilde{q}_t + \varepsilon$		$s_t \equiv -b/2$	
$(r_t, \hat{q})$ configuration	τ	printed $\tau^\star$	proved window	miscov.	$\max_t \lvert E_t\rvert$	miscov.	$\max_t \lvert E_t\rvert$
$\hat{q} \equiv +0.4$, level b	0.599	1.4	$\tau < 0.6$	0.100010	7.20	0.099890	11.60
$\hat{q} \equiv +0.4$, level b	0.8	1.4	$\tau < 0.6$	0.100050	8.70	**0.000000**	10,000
$\sup_x r_t = 10b$, off the reach	1.001	19	$\tau < 1$	**1.000000**	90,000	0.000000	10,000
$\sup_x r_t$ unbounded, off the reach	1.5	∞	$\tau < 1$	**1.000000**	90,000	0.000000	10,000
$A^+ = 4b$, $A^- = -b$	1.5	7	$\tau < 1$	0.100020	4.30	**0.000000**	10,000
level $4b$, $\hat{q} \equiv 0$	6.999	7	$\tau < 7$	0.100070	12.20	0.100050	11.60
level $4b$, $\hat{q} \equiv 0$	7.0	7	$\tau < 7$	0.100090	12.20	**0.000000**	10,000
$\sup_x r_t = 2b$, unattained	2.99	3	$\tau < 1$	0.100320	211.10	0.098010	210.50
$\sup_x r_t = 2b$, unattained	3.0	3	$\tau < 1$	**1.000000**	90,000	0.000000	10,000
$\sup_x r_t = 2b$, attained	3.0	3	$\tau < 3$	0.100000	12.20	**0.000000**	10,000
$\hat{q}_t = b/2$ on powers of 2	1.0	2	$\emptyset$	0.100009	14.30	**0.000000**	100,000
$\hat{q}_t = b/2$ on powers of 2	1.9	2	$\emptyset$	**0.110960**	84,745.70	0.000000	100,000
$\hat{q} \equiv +b/2$, no band	0	2	$\emptyset$	0.100000	0.90	**0.000000**	10,000